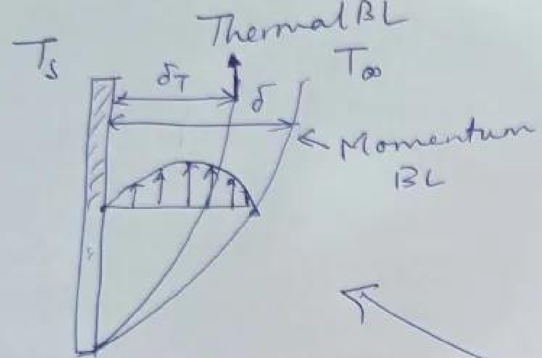
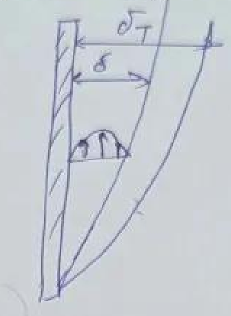
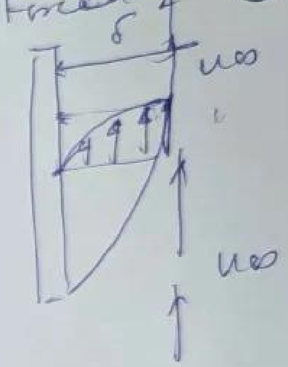


$Pr << 1 \Rightarrow \delta < \delta_T$



Forced



$Pr >> 1 \Rightarrow \delta > \delta_T$

$$\delta^2 \sim \frac{\nu x}{u_\infty}$$

$$\delta^2 \sim \nu$$

$$\delta_T^2 \sim \frac{\alpha x}{u_T} \qquad \frac{u_T}{u_\infty} \sim \frac{\delta_T}{\delta}$$

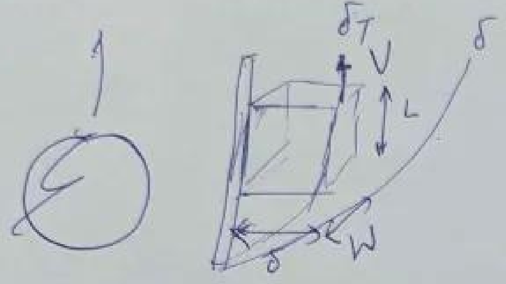
$$\left(\delta_T^2\right) \sim \frac{\alpha x}{u_\infty \delta_T/\delta}$$

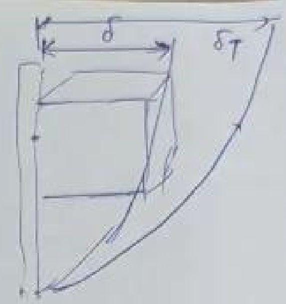
Divide

$$\frac{\delta^2}{\delta_T^2} = \frac{\dfrac{\nu \cancel{x}}{\cancel{\alpha x}\, u_\infty}}{\dfrac{\alpha \cancel{x}}{\cancel{u_\infty}\,\dfrac{\delta_T}{\delta}}}$$

$$\frac{\delta^3}{\delta_T^3} \approx \frac{\nu}{\alpha}$$

$$\Rightarrow \left(\frac{\delta}{\delta_T}\right) \sim \left(\frac{\nu}{\alpha}\right)^{1/3} = (Pr)^{1/3}$$





$$F_{Buoyancy} - F_{gravity} - F_{wall} = 0 \qquad F_{Buoyancy} - F_{gravity} = ma$$

$$F_{Buoyancy} = \rho_\infty \forall g = \rho_\infty (\delta W L) g$$

$$F_{gravity} = \rho \forall g = \rho \delta W L g$$

$$F_{Buoyancy} - F_{gravity} = (\rho_\infty - \rho)\delta W L g$$

Taylor series: $\rho(T) = \rho_\infty + \frac{d\rho}{dT}(T - T_\infty)$

Ideal gas: $p = \rho R T$

$$dp = d\rho R T + \rho R dT$$

$dp = 0 \Rightarrow$ $0 = d\rho \cancel{R} T + \rho \cancel{R} dT$

$$\frac{d\rho}{dT} = -\frac{\rho}{T}$$

$$\rho(T) = \rho_\infty - \frac{\rho}{T}(T - T_\infty)$$

$$\rho_\infty - \rho = \frac{\rho}{T}(T - T_\infty)$$

$$F_{Buoyancy} - F_{gravity} = \frac{\rho}{T}(T - T_\infty)\,\delta W L g.$$

1) $\underline{Pr >> 1}$ $\quad F_{Buoyancy} - F_{gravity} - F_{wall} = 0$

$$F_{wall} = \mu \frac{V}{\delta} L W$$

$$\mu \frac{V}{\delta} \cancel{LW} = \frac{\rho}{T}(T - T_\infty)\,\delta^2 \cancel{W} L g$$

Define $\beta \triangleq \frac{1}{T}$

$$V = g\beta(T - T_\infty)\frac{\rho}{\mu}\delta^2$$

$$V = \frac{g\beta(T - T_\infty)\delta^2}{\nu}$$

$$\delta_T^2 \sim \frac{\alpha x}{V}$$

$$\delta_T^2 \sim \frac{\alpha x \nu}{g\beta(T - T_\infty)\delta_T^2}$$

$$\frac{\delta_T^4}{x} \sim \frac{\alpha \nu}{g\beta(T - T_\infty)}$$

$$\frac{\delta_T^4}{x^4} \sim \frac{\alpha \nu}{g\beta(T - T_\infty)x^3}$$

$$\sim \frac{\alpha}{\nu}\,\frac{\nu^2}{g\beta(T - T_\infty)x^3}$$

$$\rightarrow \left(\frac{x}{\delta_T}\right)^4 = \left(\frac{\nu}{\alpha}\right)\frac{g\beta(T - T_\infty)x^3}{\nu^2}$$

$$(Pr)\,(Gr)$$

$$\text{Grashof number} \triangleq \frac{g\beta\Delta T x^3}{\nu^2}$$

$$\frac{x}{\delta_T} \sim (Pr)^{1/4}(Gr)^{1/4}$$

$$\delta_T^2 \sim \frac{\alpha x}{V}$$

$$\delta_T^4 \sim \frac{\alpha^2 x^2}{V^2}$$

$$\sim \frac{\alpha^2 x^{\cancel{2}}}{\beta \Delta T \cancel{x} g}$$

$$\sim \frac{\alpha^2 x}{\beta \Delta T g}$$

$$\frac{\delta_T^4}{x} \sim \frac{\alpha^2}{\beta \Delta T g}$$

$$\frac{\delta_T^2}{x^4} \sim \alpha^2 \frac{1}{\beta \Delta T g x^3}$$

$$\sim \frac{\alpha^2}{\nu^2} \frac{\nu^2}{\beta \Delta T g x^3}$$

$$\left(\frac{x}{\delta_T}\right)^4 \sim \left(\frac{\nu}{\alpha}\right)^2 \left(\frac{\beta \Delta T g x^3}{\nu^2}\right)$$

$$\left(\frac{x}{\delta_T}\right)^4 \sim (Pr)^2 (Gr)$$

$$\frac{x}{\delta_T} \sim (Pr)^{1/2} (Gr)^{1/4}$$

$$Nu = \frac{h\cancel{x}}{k}$$

$$= \frac{k}{\delta_T} \frac{x}{k}$$

$$= \frac{x}{\delta_T}$$

$$\boxed{Nu \sim (Pr)^{1/2} (Gr)^{1/4}}$$

For intermediate Pr

$$Nu \sim \frac{A (Pr)^{1/2} (Gr)^{1/4}}{1 + B(Pr)^{1/4}(\;)}$$

$$Nu = \frac{h x}{k}$$

$$q = h(T - T_\infty)$$

$$Nu = \frac{hx}{k}$$

$$= \frac{\cancel{k}}{\delta_T} \frac{x}{\cancel{k}}$$

$$= \frac{x}{\delta_T}$$

$$\approx (Pr)^{1/4} (Gr)^{1/4}$$

(2) $\underline{Pr << 1}$ $F_{Buoyancy} - f_{gravity} = ma$

$$m = \rho W L \delta$$

$$a \approx \frac{V}{t} \qquad t = \frac{x}{V}$$

$$\approx \frac{V^2}{x}$$

$$\frac{\cancel{\rho}}{T} \Delta T \cancel{\delta} \cancel{W} \cancel{L} g \approx \frac{V^2}{x} \cancel{\rho} \cancel{W} \cancel{L} \cancel{\delta}$$

$$V^2 = \frac{\Delta T}{T} g x$$

$$\approx \beta \Delta T g x .$$